

1. RUN CONFIGURATION

Frames Processed	Duration	Processing FPS	Avg Processing	Max Processing
141	0.92 s	153.8	3.43 ms	6.97 ms

2. TRACKING PERFORMANCE

Acquisition Time	Average Tracking Error	Maximum Tracking Error
0.03 s	3.66 px	6.08 px
<= 2 s		<= 10 px
RMSE Tracking Error	Centroid Error Avg/Max	Lock Retention
3.83 px	3.66 / 6.08 px	100.00 %
	ground truth only	
Target Loss	Re-acquisition Avg/Max	Successful Re-acq.
0.00 %	—	0
< 5 %	<= 1 s	0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.03	<= 2.0 s	PASS
Tracking Error (Max)	6.08	<= 10 px	PASS
Center Error (Final)	200.20	<= 10 px	FAIL
Target Loss	0.00 %	< 5%	PASS
Re-acquisition Time	—	<= 1.0 s	PENDING
Processing Speed	153.8 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: FAIL

Source: benchmark video: synthetic_fsoc_beacon.mp4 | Frames: 141 | FPS: 153.8